

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Folate Forms - Total & Individual - Washed RBCs (FFMR_L)

Data File: FFMR_L.xpt

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Component Description

Folate belongs to the group of water-soluble B vitamins that occur naturally in food. Prolonged folate deficiency leads to megaloblastic anemia. Low folate status has been causally linked to an increased risk in women of reproductive age to have an offspring with neural tube defects. Low folate status also increases plasma homocysteine levels, a potential risk factor for chronic diseases, such as cardiovascular disease or cognitive function. Potential roles of folate and other B vitamins in modulating the risk for diseases (e.g., heart disease, cancer, and cognitive impairment) are under investigation. While serum folate is an indicator of recent intake, red blood cell (RBC) folate is an indicator of long-term status.

These data will be used to estimate deficiencies and other health effects of specific nutrients in the population and subgroup, to provide population reference data, and to estimate the contribution of diet, supplements, and other factors to the levels of nutrients. Data will be used in research to further define nutrient requirements as well as optimal levels for disease prevention and health promotion. Data will also be used to evaluate the effect of changes in nutrition and public health policies, including welfare reform legislation, food fortification policy, and child nutrition programs on the nutritional status of the U.S. population.

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

RBC folate status can be determined directly by measuring folate forms in washed RBCs (Stamm et al., 2018) or indirectly by measuring folate forms in a whole blood (WB) lysate (Fazili et al., 2004; Fazili et al., 2005). The direct measurement requires the addition of exogenous γ -glutamyl hydrolase (exo-GGH) to deconjugate folate polyglutamates to monoglutamates (Stamm et al., 2018). It also requires the measurement of hemoglobin (Hb) in the RBC lysate to correct for residual moisture (~20%) in the washed RBCs. By using the mean corpuscular hemoglobin content (MCHC), RBC folate can be calculated. The indirect measurement can either utilize the endogenous plasma-based γ -glutamyl hydrolase (no addition of enzyme; 4 h incubation at 37°C) or preferably, for faster deconjugation, use the addition of exo-GGH (30 min incubation at ambient temperature [+15°C to +30°C]). To calculate RBC folate, it requires the measurement of serum folate to correct for the serum folate contribution to WB folate and hematocrit (Hct) to normalize to the proportion of packed red cells.

For this NHANES component, RBC folate status was determined directly by measuring folate forms in washed RBCs.

Five folate forms, 5-methylTHF, pteroylglutamic acid (Folic Acid, PGA), tetrahydrofolate (THF), 5-formyl-tetrahydrofolic acid (5-formylTHF), 5,10-methenyltetrahydrofolic acid (5,10-methenylTHF), and one oxidation product of 5-methylTHF called MeFox (pyrazino-s-triazine

derivative of 4- α -hydroxy-5-methylTHF) are measured by isotope-dilution high performance liquid chromatography coupled to tandem mass spectrometry (HPLC-MS/MS). The current method is a modification of previously published methods (Fazili et al., 2004; Fazili, et al., 2005) to add the measurement of an oxidation product (Fazili et al., 2013;143:108-113) and to scale down the amount of specimen needed and increase the sample throughput (Fazili et al., 2013;405:4549-60). The method distinguishes 5-formylTHF from 10-formylTHF on the basis of mass transitions; however, during HPLC-MS/MS in the acidic mobile phase, 10-formylTHF converts within minutes to 5,10-methenylTHF and trace amounts of 5-formylTHF, THF, and 10-formyl-folic acid. Thus, this method is not capable of quantifying 10-formylTHF; formylated folates are quantified as 5-formylTHF and 5,10-methenylTHF, which are stable under these conditions (Pfeiffer et al., 2004).

The assay is performed by combining specimen (150 μ L of WB lysate or RBC lysate) with an internal standard mixture containing exo-GGH enzyme (5 μ g per mL of WB lysate or RBC lysate) and incubating at ambient temperature (+15°C to +30°C) for 30 min to deconjugate folate polyglutamates to monoglutamates prior to folate extraction. Ammonium formate buffer (1%) is added to the samples and extraction and clean-up is performed by automated 96-probe solid phase extraction (SPE) using 96-well phenyl SPE plates which takes ~1 h for a 96-well plate. Folate forms are separated within 4 min using isocratic mobile phase conditions and measured by HPLC-MS/MS (7 min to next injection). Quantitation is based on peak area ratios interpolated against a five-point aqueous linear calibration curve using 1/x² weighting.

RBC folate form concentrations (nmol/L) measured in an RBC lysate are calculated as RBC folate form = (RBC lysate folate form / Hb) * MCHC.

Refer to the Laboratory Method Files section for a detailed description on the laboratory methods used.

Laboratory Method Files

[Folate Forms - Total & Individual – Washed RBCs](#) (January 2026)

Laboratory Quality Assurance and Monitoring

Washed red blood cell specimens are processed, stored, and shipped to the Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (–30°C) conditions until they are shipped to the National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendments. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured QA evaluation during unscheduled visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected during “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all the contract laboratories, which outlined the use of Westgard rules (Westgard, et al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences’ QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill, et al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

RBC Folate Forms for NHANES 2021-2023

In NHANES 2021-2023, washed red blood cells (RBC) were diluted approximately 1+1 with saline and then 1+10 with ascorbic acid solution to generate RBC lysates. Several folate forms were measured in RBC lysates by isotope-dilution high-performance liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS). Instrument-measured lysate concentrations were converted to concentrations expressed in nmol/L RBC by dividing by the Hb content measured in the same aliquot and multiplying by the participant-specific MCHC to account for the ‘dilution’ effect from washed RBC to lysate (Table 1, LBXRF1SI to LBXRF6SI). For individual RBC folate forms with lysate concentrations below the limit of detection (LOD), an imputed value equal to the LOD divided by the square root of 2 was used prior to calculating the RBC concentration (nmol/L RBC). RBC folate form concentrations were not calculated when the Hb or MCHC value was missing. Total RBC folate (LBDRF7SI) was calculated as the sum of individual folate forms LBXRF1SI through LBXRF5SI. LBXRF6SI was

not included in the total folate calculation, because this folate form is not biologically active and due to evidence that it may already be present in vivo (Pfeiffer, et al., 2015). RBC total folate concentration was not calculated when one or more individual RBC folate form results were missing.

Table 1. Folate forms measured by LC-MS/MS

Analyte	Abbreviation	Variable Name
5-Methyltetrahydrofolate	5-methylTHF	LBXRF1SI
Pteroylglutamic acid (Folic Acid)	PGA	LBXRF2SI
5-Formyltetrahydrofolate	5-formylTHF	LBXRF3SI
Tetrahydrofolate	THF	LBXRF4SI
5,10-Methenyltetrahydrofolate	5,10-methenylTHF	LBXRF5SI
Pyrazino-s-triazine derivative of 4-a-hydroxy-5-methyltetrahydrofolate	MeFox	LBXRF6SI
Total washed RBC folate (sum of biologically active folate forms)	tFOL	LBDRF7SI

Demographic and Other Related Variables

The analysis of NHANES laboratory data may require additional demographic variables. The [NHANES August 2021- August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews.

This laboratory data file can be linked to the Demographics file and other NHANES data files in the August 2021-August 2023 cycle using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The lower limit of detection (LLOD) for the five folate forms and the oxidation product of 5-methylTHF applies to instrument measurements performed on RBC lysates, not to the reported concentrations of folate forms in RBC (nmol/L RBC) that are presented in the data. Reported RBC folate concentrations are calculated from the instrument-measured lysate values using each participant's Hb and MCHC. As a result, even when the underlying instrument measurement is below the detection limit, the reported values expressed in nmol/L RBC will be larger than the instrument LLOD values. Because values below the LLOD were addressed during the calculation of RBC concentrations from lysate instrument measurements, data analysts do not need to apply the LLOD values in subsequent analyses. However, an indicator variable is provided. The variable name ending in "LC" (ex., LBDRF1LC) indicates whether the instrument-measured lysate value was below the LLOD (0 = at or above LLOD; 1 = below LLOD). Measurements indicated as below the LLOD retain the increased uncertainty associated with instrument results below the detection limit, even when the reported RBC concentration is greater than the LLOD. The corresponding variable prefixed LBX (ex., LBXRF1SI) reports the calculated concentration in nmol/L RBC.

Total folate was calculated as the sum of individual folate forms (RF1–RF5); therefore, a lower limit of detection does not apply to total folate.

References

- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. *Statist. Med.* (2008) 27(20):4094-40106.

- Fazili Z, Pfeiffer CM, Zhang M, Jain R. Erythrocyte folate extraction and quantitative determination by liquid chromatography-tandem mass spectrometry: comparison of results with microbiologic assay. Clin Chem 2005; 51:2318–25.
- Fazili Z, Pfeiffer CM. Accounting for an isobaric interference allows correct determination of folate vitamers in serum by isotope dilution-liquid chromatography-tandem MS. J Nutr. 2013;143:108–113.
- Fazili Z, Pfeiffer CM. Measurement of folates in serum and conventionally prepared whole blood lysates: Application of an automated 96-well plate isotope-dilution tandem mass spectrometry method. Clin Chem. 2004;50:2378–81.
- Fazili Z, Whitehead RD Jr, Paladugula N, Pfeiffer CM. A high-throughput LC-MS/MS method suitable for population biomonitoring measures five serum folate vitamers and one oxidation product. Anal Bioanal Chem. 2013;405:4549–60
- Pfeiffer C, Sternberg M, Fazili M, Lacher D, Zhang M, Johnson C, Hammer H, Baily R, Rader J, Yamini S, Berry RJ, Yetley E. British Journal of Nutrition (2015) 113:1965:1977.
- Pfeiffer CM, Fazili Z, McCoy L, Zhang M, and Gunter EW. Determination of folate vitamers in human serum by stable-isotope dilution tandem mass spectrometry and comparison to radioassay and microbiologic assay. Clin Chem. 2004;50:423–32.
- Stamm R, Fazili Z, Pfeiffer CM. Addition of exogenous γ -glutamyl hydrolase eliminates the need for lengthy incubation of whole blood lysate for quantitation of folate vitamers by high-performance liquid chromatography-tandem mass spectrometry. Curr Dev Nutr. 2018;2:1–9.
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493–501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBDRF7SI - Total Folate, RBC (nmol/L)

Variable Name: LBDRF7SI
SAS Label: Total Folate, RBC (nmol/L)
English Text: Total folate, washed Red Blood Cells (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
125 to 6360	Range of Values	7190	7190	
.	Missing	878	8068	

LBXRF1SI - 5-Methyl-tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF1SI
SAS Label: 5-Methyl-tetrahydrofolate, RBC (nmol/L)
English Text: 5-Methyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
28 to 5960	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF1LC - 5-Methyl-tetrahydrofolate, RBC cmt

Variable Name: LBDRF1LC
SAS Label: 5-Methyl-tetrahydrofolate, RBC cmt
English Text: 5-Methyl-tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7190	7190	
1	Below lower detection limit	0	7190	
.	Missing	878	8068	

LBXRF2SI - Folic acid, RBC (nmol/L)

Variable Name: LBXRF2SI
SAS Label: Folic acid, RBC (nmol/L)
English Text: Folic acid, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.664 to 72.5	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF2LC - Folic acid, RBC cmt

Variable Name: LBDRF2LC**SAS Label:** Folic acid, RBC cmt**English Text:** Folic acid, RBC comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	3230	3230	
1	Below lower detection limit	3960	7190	
.	Missing	878	8068	

LBXRF3SI - 5-Formyl-tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF3SI
SAS Label: 5-Formyl-tetrahydrofolate, RBC (nmol/L)
English Text: 5-Formyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.92 to 45.5	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF3LC - 5-Formyl-tetrahydrofolate, RBC cmt

Variable Name: LBDRF3LC
SAS Label: 5-Formyl-tetrahydrofolate, RBC cmt
English Text: 5-Formyl-tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	2985	2985	
1	Below lower detection limit	4205	7190	
.	Missing	878	8068	

LBXRF4SI - Tetrahydrofolate, RBC (nmol/L)

Variable Name: LBXRF4SI
SAS Label: Tetrahydrofolate, RBC (nmol/L)
English Text: Tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.84 to 1360	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF4LC - Tetrahydrofolate, RBC cmt

Variable Name: LBDRF4LC
SAS Label: Tetrahydrofolate, RBC cmt
English Text: Tetrahydrofolate, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7101	7101	
1	Below lower detection limit	89	7190	
.	Missing	878	8068	

LBXRF5SI - 5,10-Methenyl-tetrahydrofolate (nmol/L)

Variable Name: LBXRF5SI
SAS Label: 5,10-Methenyl-tetrahydrofolate (nmol/L)
English Text: 5,10-Methenyl-tetrahydrofolate, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.56 to 1020	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF5LC - 5,10-Methenyl-tetrahydrofolate, RBC cmt**Variable Name:** LBDRF5LC**SAS Label:** 5,10-Methenyl-tetrahydrofolate, RBC cmt**English Text:** 5,10-Methenyl-tetrahydrofolate, RBC comment code**Target:** Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	5291	5291	
1	Below lower detection limit	1899	7190	
.	Missing	878	8068	

LBXRF6SI - MeFox oxidation product, RBC (nmol/L)

Variable Name: LBXRF6SI
SAS Label: MeFox oxidation product, RBC (nmol/L)
English Text: MeFox oxidation product, RBC (nmol/L)
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7.57 to 1630	Range of Values	7190	7190	
.	Missing	878	8068	

LBDRF6LC - MeFox oxidation product, RBC cmt

Variable Name: LBDRF6LC
SAS Label: MeFox oxidation product, RBC cmt
English Text: MeFox oxidation product, RBC comment code
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above the detection limit	7190	7190	
1	Below lower detection limit	0	7190	
.	Missing	878	8068	